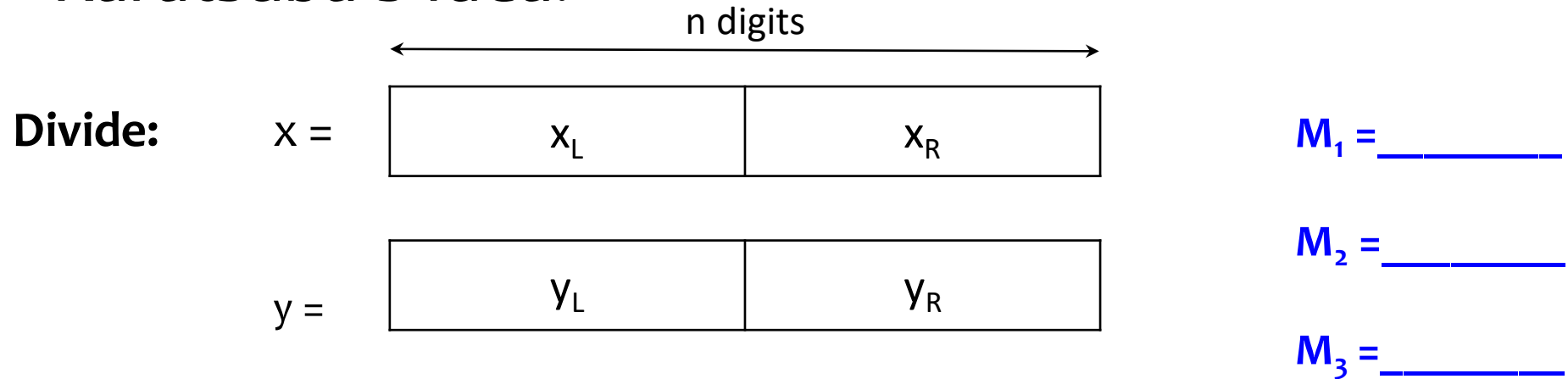


Lecture 3 Handout: Divide and Conquer 2

Karatsuba's idea!



Conquer: $x \cdot y = \underline{\hspace{1cm}} \cdot 10^n + \underline{\hspace{1cm}} \cdot 10^{n/2} + \underline{\hspace{1cm}}$

||

$= M_2 \cdot 10^n + (\underline{\hspace{2cm}}) \cdot 10^{n/2} + M_3$

Next: Analyze the runtime of Karatsuba with the Master Theorem

Solving the recurrence

The Master Theorem

Let $T(n)$ be the following recurrence, where $k \geq 1$, $b > 1$, $d \geq 0$, $w \geq 0$ are constants (they do not depend on n):

$$T(n) = kT(n/b) + O(n^d \log^w n)$$

Then:

$$T(n) = \begin{cases} O(n^d \log^w n) & \text{if } k < b^d \\ O(n^d \log^{w+1} n) & \text{if } k = b^d \\ O(n^{\log_b k}) & \text{if } k > b^d \end{cases}$$

$$T(n) = T(n/__) + O(__)$$

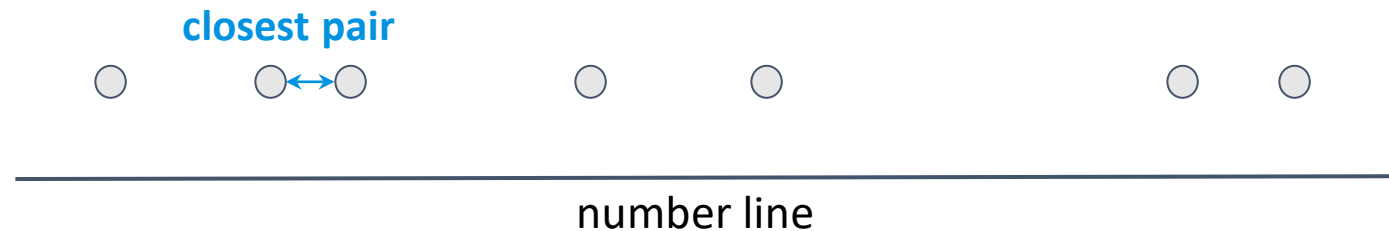
Conclusion: Karatsuba's Algorithm takes $O(\text{---})$ time.

Warm-up: Closest Pair of Points in 1-D

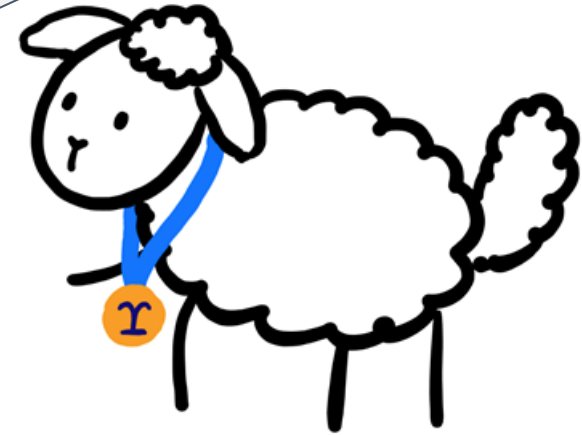
* **Problem:** Given n real numbers $x_1, x_2, \dots, x_n$, find $i \neq j$ with the smallest $|x_i - x_j|$.

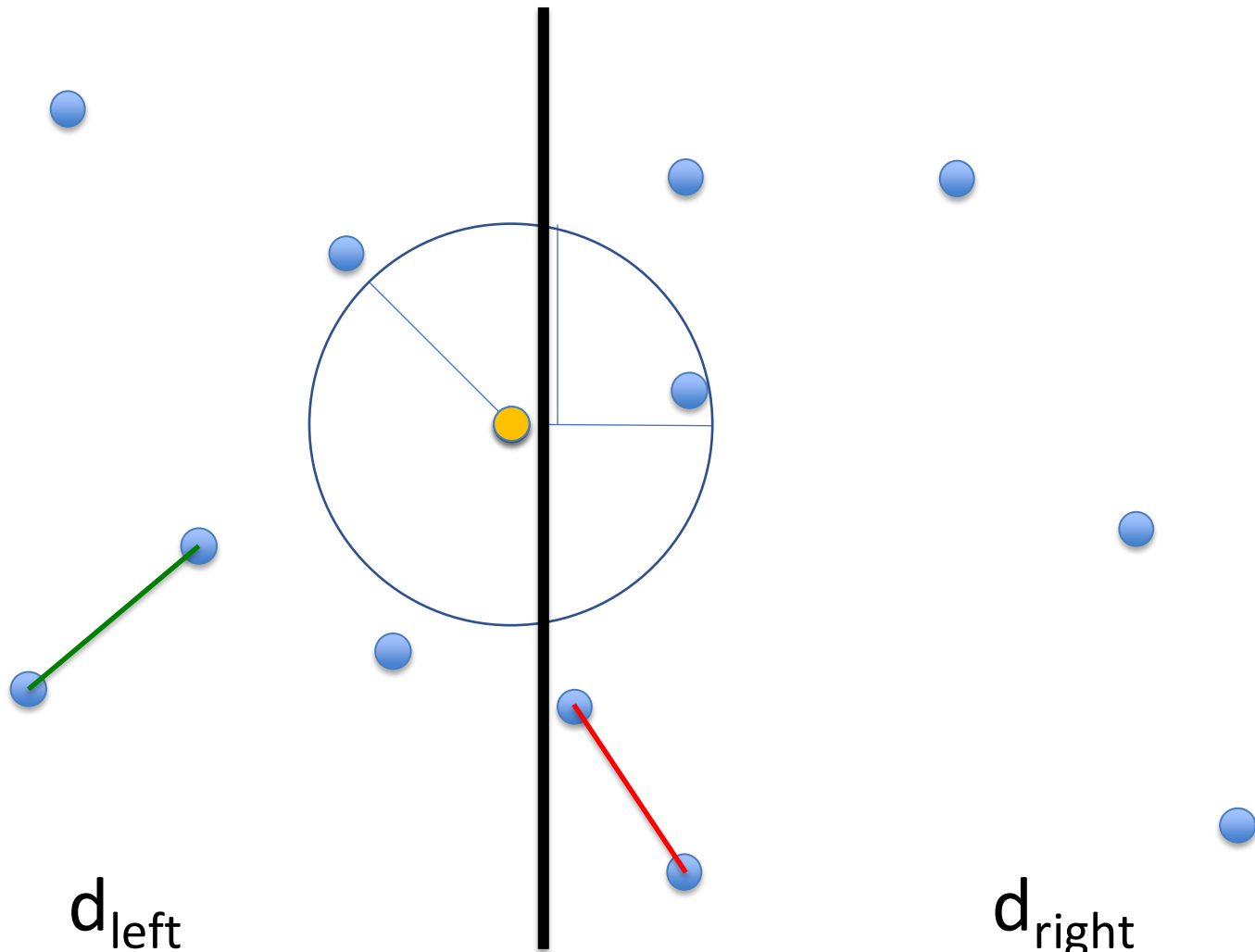
* **Solution:**

* **Runtime:**



No need to use divide-and-conquer! Just run my new algorithm! Iterate through the sorted list X and compute the distance between adjacent points. Then do the same for the list Y. Then take the minimum distance we found!

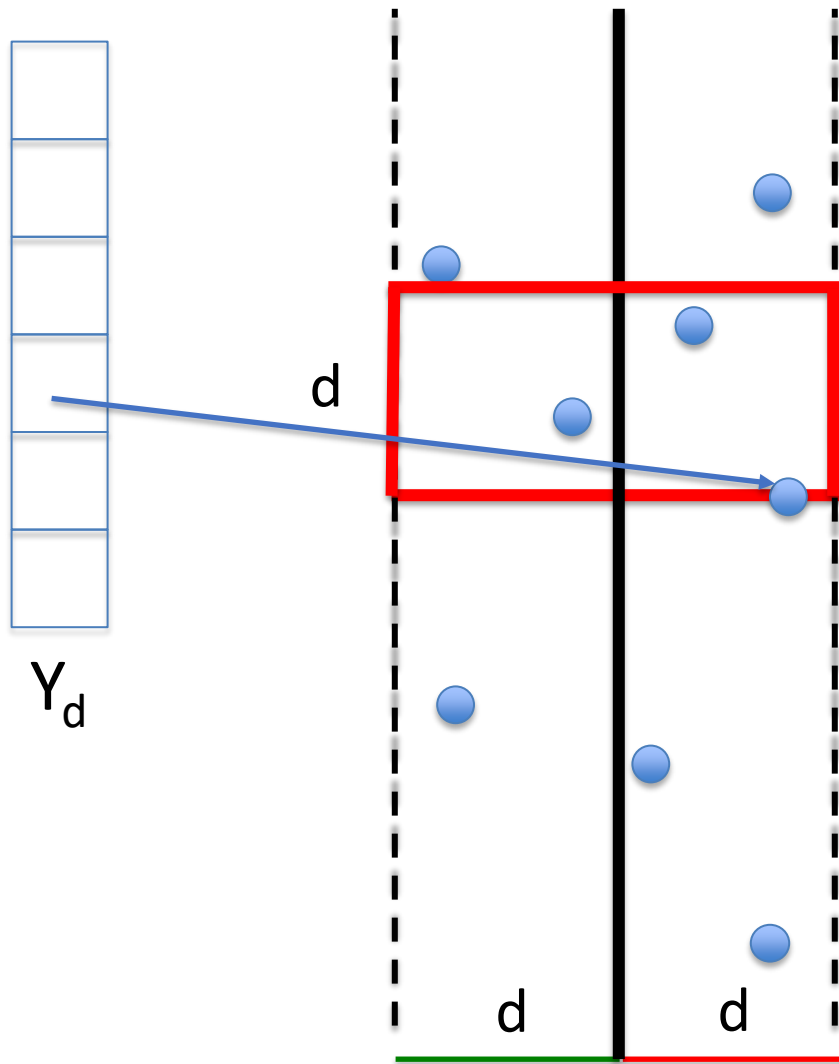




Good news! (?)



$$d = \min(d_{\text{left}}, d_{\text{right}})$$



Claim: There can't be too many points in this red ____-by-____ rectangle? Why? How many points could there be?



Algorithm Recap

X = array of points sorted by x-coordinate

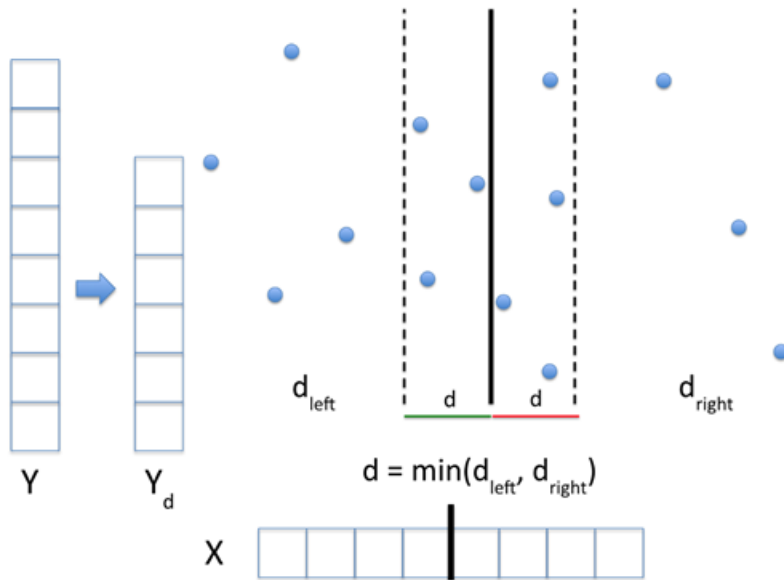
Y = array of points sorted by y-coordinate

ClosestPair(X, Y):

1. Divide points into left and right, and create $X_{\text{left}}, X_{\text{right}}, Y_{\text{left}}, Y_{\text{right}}$
2. $d_{\text{left}} = \text{ClosestPair}(X_{\text{left}}, Y_{\text{left}})$
3. $d_{\text{right}} = \text{ClosestPair}(X_{\text{right}}, Y_{\text{right}})$
4. $d = \min(d_{\text{left}}, d_{\text{right}})$
5. Y_d = array of points of distance $\leq ___$ from middle line sorted by y-coord
6. For each point p in Y_d : compare p to the **next $___$ points** in Y_d .
7. d_{next} = minimum distance found in step 6.
8. Return $\min(d, d_{\text{next}})$

Total Running Time

- Sort once by x-coordinate: $O(n \log n)$
- Sort once by y-coordinate: $O(n \log n)$
- Recursive algorithm: $T(n) = __T(n/__) + O(__)$



Discovered in 1976 by
Jon Louis Bentley and Michael Ian Shamos

